



Topic to cover :

① Async vs sync

- ① Core Problem: Waiting
Backend often does slow I/O.
- Database calls
 - API calls
 - File reads
 - Network request

If your server wait doing nothing, it waste time.

- ② Sync (Blocking) Function
- ```
import time
def get_data():
 time.sleep(3) # Block everything
 return "data"
```

If one request comes :

- Server wait 3 second
- During that time, it can't handle others efficiently

Analogy:

one waiter

Takes one order

Waits in kitchen

Other customer keep waiting



### ③ Async (Non-blocking) Function

```
import asyncio
async def get_data():
 await asyncio.sleep(3) # doesn't block server
 return "data"
```

Now:

- While waiting, server can handle other request.
- Much better concurrency

Analogy:

- One waiter
- Place order
- While food is cooking, serve other tables

### ④ FastAPI + Async

FastAPI support both:

```
@app.get("/sync")
```

```
def sync_endpoint():
```

```
 time.sleep(3)
```

```
 return {"message": "sync done"}
```

```
@app.get("/async")
```

```
async def async_endpoint():
```

```
 await asyncio.sleep(3)
```

```
 return {"message": "async done"}
```

- /sync block worker
- /async allow concurrency



### Topic Covered:

- ① Background Task
- ② Long Jobs
- ③ Not Blocking Users

### ① The Problem: Long Running Task

Some task take time:

- Sending emails
- Processing Files
- Generating reports
- calling slow APIs
- ML Prediction

If you do this inside the request:

- User Wait
- API feel slow
- Server thread get blocked

### ② FastAPI Background Task

From FastAPI import BackgroundTask

You can:

- return response immediately
- Run function in background

### ③ Simple Example: send Email



```
from Fastapi import BackgroundTasks
import time
def send_email(email: str):
 time.sleep(5) # simulate slow work
 print(f"Email sent to {email}")
```

Endpoint:

```
@router.post("/send-email")
def send_email_endpoint(email: str,
 background_task: BackgroundTasks):
 background_task.add(send_email, email)
 return {"message": "Email will be
 sent in background."}
```

(4) What Happens?

1. Client calls /send-email
2. API immediately return response
3. send\_email() run in background
4. Server is free to handle other request.

User doesn't wait 5 seconds.

(5) Real Analogy

Restaurant:

- ① You place order
- ① You get token immediately.
- ① Kitchen prepare food in background
- ① You don't block the counters.

⑥ When to use Background Task?

- |                         |                    |
|-------------------------|--------------------|
| ① Sending email         | Not good for:      |
| ① Logging               | ① ML training      |
| ① Notification          | ① video processing |
| ① Small File Processing | ① Big data         |